

Batch Normalization ✓

✓ Normalization

$$\underline{\text{minmax scale}} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (0, 1)$$

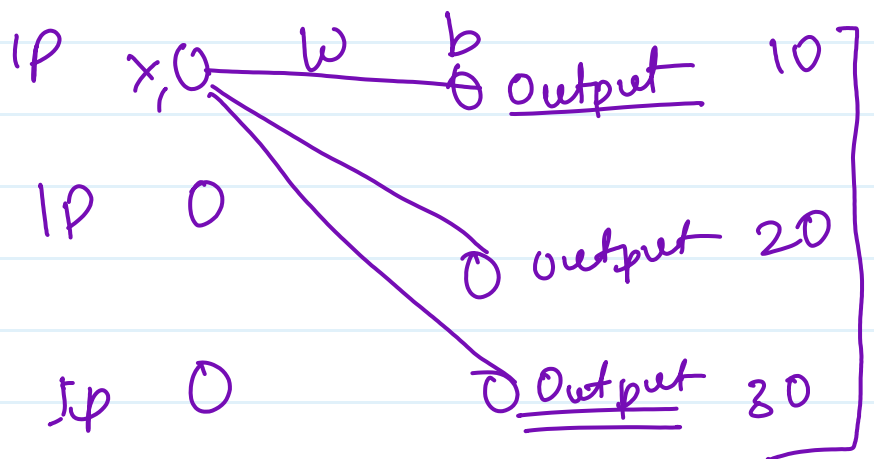
Data $\frac{SC}{NR}$ burning

Norm. $IP \rightarrow HL \rightarrow HL \rightarrow HL \rightarrow OP$

$$z = wx + b$$

$$\sigma = \frac{1}{1 + e^2}$$

= value



→ 0

→ 0

→ 0

f_1	f_2	f_3	f_3
1			
⋮			
1			

2DNN

$$\begin{pmatrix} -1 & +1 \\ 0 & 1 \end{pmatrix}$$

0 255



4x4

10		0	9
	100		
	135	255	
			105

BW

— 0 — 1

255

$$\frac{0}{255} = 0$$

$$\frac{100}{255} \Rightarrow 0.39$$

$$\frac{255}{255} = 1$$

HL_0	μ	σ^2	σ
0 10	$\frac{100}{4}$	$(10-25)^2$	$\sqrt{125}$
0 20		$(20-25)^2$	$\boxed{11.18}$
0 30	$\boxed{25}$	$(30-25)^2$	
0 40		$(40-25)^2$	
		$\Rightarrow \boxed{125}$	

— formula — $\hat{x}_i = \frac{x_i - \mu}{(\sigma + \epsilon)}$

$$\epsilon = 0.0001$$

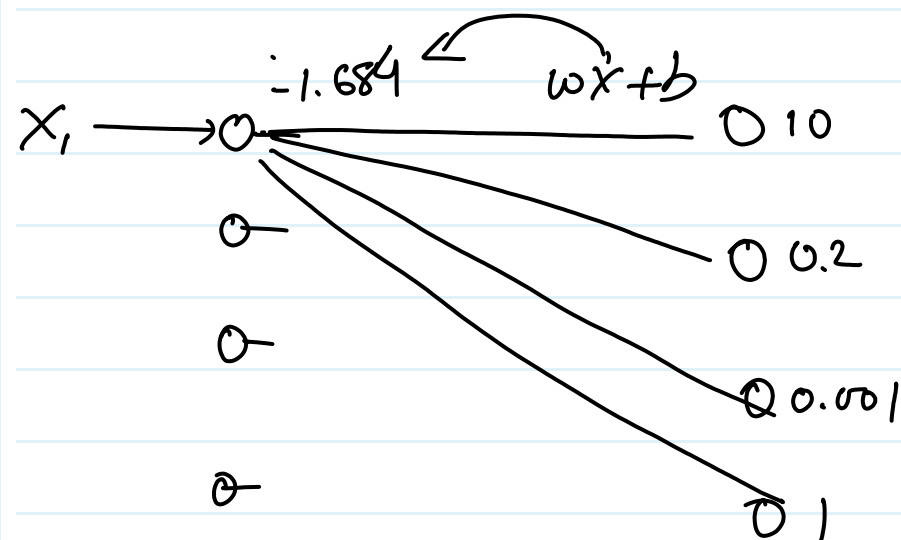
X	$X - \mu$	σ	$\hat{x}_i$ Normalization
10	-15	-15/11.18	-1.342
20	-5	-5/11.18	-0.447
30	5	5/11.18	0.447
40	15	15/11.18	1.342

$\hat{x}_i$	$\gamma \cdot \hat{x}_i$	β	y
-1.342	$-1.342 \times 2 = -2.684$	1	-1.684
-0.447	$-0.447 \times 2 = -0.894$	1	0.106
0.447	$0.447 \times 2 = 0.894$	1	1.894
1.342	$1.342 \times 2 = 2.684$	1	3.684

scale & shift (γ and β)

$$\gamma = 2, \beta = 1$$

$$y_i = \gamma \cdot \hat{x}_i + \beta \quad (-1 \quad +1)$$



Early stopping

→ when you train NN too long. It start memorizing the training data instead of learning pattern.

Early stopping = stop training the model at right moment -

When it performing best on unseen data, before it start overfit.

Epoch	Train loss	val loss	Action
1	0.90	0.88	improving - cont.
2	0.75	0.74	improving - cont.
3	0.60	0.59	improving - cont.
✓ 4	0.60	0.45	✓ Best so far - cont.
5	0.35	0.65	val going up - Patience 1/3
6	0.28	0.72	worse - patience 2/3
7.	0.20	0.80	stop on epoch 4
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check point -

Saving a snapshot of your model at a specific point during training so we can resume or use it later

Game start - level - 0.

level - 1

level - 2 ✓

✓ level - 3 Running

level - 4

level - 5 stuck